TWO NONSEPARABLE GENERALIZATIONS OF THE $\mathbb{G}_0$ DICHOTOMY

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Abstract. We give classical proofs of both the generalization of the uniform $\mathbb{G}_0$ dichotomy to sequences of digraphs indexed by $\kappa^{\mathbb{N}}$ and the generalization of the $\mathbb{G}_0$ dichotomy to κ-Souslin digraphs.

Introduction

Suppose that κ is an infinite cardinal and X is a Hausdorff space. Endow κ with the discrete topology and $\kappa^{\mathbb{N}}$ with the product topology. For all $n \in \mathbb{N}$ and $s \in \kappa^n$, set $|s| = n$. For all $\sigma \in \kappa^{\leq\mathbb{N}}$, let $s \sqsubseteq \sigma$ indicate that $\forall i < |s|\ s(i) = \sigma(i)$. Define $\mathcal{N}_s = \{\sigma \in \kappa^{\mathbb{N}} \mid s \sqsubseteq \sigma\}$. We write $\forall^*\sigma \in \kappa^{\mathbb{N}}\ \phi(\sigma)$ to indicate that $\{\sigma \in \kappa^{\mathbb{N}} \mid \phi(\sigma)\}$ is comeager. The *projection* function from $\prod_{i\in I} X_i$ to X_i is given by $\mathrm{proj}_{X_i}(x) = x(i)$. The *horizontal and vertical sections* of a set $R \subseteq X \times Y$ are given by $R^y = \{x \in X \mid x\ R\ y\}$ and $R_x = \{y \in Y \mid x\ R\ y\}$. Given a function $\pi\colon R \subseteq X \times Y \to Z$, define $\pi_x\colon R_x \to Z$ by $\pi_x(y) = \pi(x, y)$.

We say that $\phi\colon C \subseteq \kappa^{\mathbb{N}} \times \mathbb{N}^{\mathbb{N}} \to \kappa^{\mathbb{N}} \times X$ is *parameter preserving* if $\mathrm{proj}_{\kappa^{\mathbb{N}}} \circ \phi = \mathrm{proj}_{\kappa^{\mathbb{N}}} \restriction C$, and $A \subseteq \kappa^{\mathbb{N}} \times X$ has *uniformly analytic (vertical) sections* if there are a closed set $C \subseteq \kappa^{\mathbb{N}} \times \mathbb{N}^{\mathbb{N}}$ and a continuous parameter-preserving function $\phi\colon C \to \kappa^{\mathbb{N}} \times X$ such that $A = \phi(C)$.

Recall that a sequence $(X_i)_{i\in I}$ of subsets of a topological space is *open separated* if there are pairwise disjoint open sets U_i such that $X_i \subseteq U_i$ for all $i \in I$. A subset of a topological space is *quasi-Borel* if it is in the closure of the open sets under complements, countable unions, and open-separated unions. A function $c\colon \kappa^{\mathbb{N}} \times X \to \mathbb{N}$ is *quasi-Borel* if $c^{-1}(\{n\})$ is quasi-Borel for all $n \in \mathbb{N}$.

A digraph on X is an irreflexive set $G \subseteq X \times X$. A *homomorphism* from a digraph G on X to a digraph H on Y is a function $\pi\colon X \to Y$ such that $(\pi \times \pi)(G) \subseteq H$. A *$Y$-coloring* of G is a homomorphism $c\colon X \to Y$ from G to inequality. Fix $\mathbb{s} \in \prod_{n\in\mathbb{N}} 2^n$ such that $\mathbb{s}(\mathbb{N})$ is dense in $2^{<\mathbb{N}}$ and define $\mathbb{G}_0 = \bigcup_{n\in\mathbb{N}}\{(\mathbb{s}_n \frown (i) \frown c)_{i<2} \mid c \in 2^{\mathbb{N}}\}$.

The uniform $\mathbb{G}_0$ dichotomy for digraphs indexed by $\mathbb{N}^{\mathbb{N}}$ is a straightforward corollary of the $\mathbb{G}_0$ dichotomy itself (see [Mil12, Theorem 31]). Here we show that the following generalization to digraphs indexed

by $\kappa^{\mathbb{N}}$ follows from the observation that the terminal independent sets arising in the classical proof of the $\mathbb{G}_0$ dichotomy can be grouped into countably many open-separated families, preserving both definability and the number of colors:

Theorem 1. *Suppose that κ is an infinite cardinal, X is a Hausdorff space, and $G \subseteq \kappa^{\mathbb{N}} \times (X \times X)$ has uniformly analytic sections that are digraphs on X. Then exactly one of the following holds:*

1. *There is a quasi-Borel function $c\colon \kappa^{\mathbb{N}} \times X \to \mathbb{N}$ such that c_σ is a coloring of G_σ for all $\sigma \in \kappa^{\mathbb{N}}$.*
2. *There is a continuous homomorphism $\pi\colon 2^{\mathbb{N}} \to X$ from $\mathbb{G}_0$ to G_σ for some $\sigma \in \kappa^{\mathbb{N}}$.*

A subset of a topological space is *κ-Souslin* if it is a continuous image of a closed subset of $\kappa^{\mathbb{N}}$. A subset of a topological space is *κ^+-Borel* if it belongs to the smallest family containing the open sets and closed under complements and unions of cardinality at most κ. A function $c\colon X \to \kappa$ is *κ^+-Borel* if $c^{-1}(\{\alpha\})$ is κ^+-Borel for all $\alpha < \kappa$.

By combining Theorem 1 with a Baire category argument, we obtain a classical proof of the following generalization of the $\mathbb{G}_0$ dichotomy:

Theorem 2 (Kanovei [Kan97])**.** *Suppose that κ is an infinite cardinal, X is a Hausdorff space, and G is a κ-Souslin digraph on X. Then at least one of the following holds:*

1. *There is a κ^+-Borel coloring $c\colon X \to \kappa$ of G.*
2. *There is a continuous homomorphism from $\mathbb{G}_0$ to G.*

In §1, we note closure properties of sets with uniformly analytic sections. In §2, we show that the horizontal sections of quasi-Borel sets have the Baire property. In §3, we generalize Lusin's separation theorem to sets with uniformly analytic sections. In §4, we prove Theorem 1. And in §5, we prove Theorem 2.

1. Pointclasses

We direct the reader to [Kec95, §14] for the most basic facts concerning analytic sets.

Proposition 1.1. *Suppose that κ is an aleph and X is a Hausdorff space. Then the subsets of $\kappa^{\mathbb{N}} \times X$ with uniformly analytic sections are closed under countable intersections and countable unions.*

Proof. By the proof of the analogous fact for analytic sets. ⊠

Proposition 1.2. *Suppose that κ is an aleph, X and Y are Hausdorff spaces, $A \subseteq \kappa^{\mathbb{N}} \times X$ and $B \subseteq \kappa^{\mathbb{N}} \times Y$ have uniformly analytic sections,*

and $f\colon A \to \kappa^{\mathbb{N}} \times Y$ is continuous and parameter preserving. Then $f(A)$ and $f^{-1}(B)$ have uniformly analytic sections.

Proof. By the proof of the analogous fact for analytic sets. ⊠

Proposition 1.3. *Suppose that κ is an aleph, X is a Hausdorff space, and $A_s \subseteq \mathcal{N}_s \times X$ has uniformly analytic sections for all $s \in \kappa^{<\mathbb{N}}$. Then $\bigcup_{s\in\kappa^{<\mathbb{N}}} A_s$ has uniformly analytic sections.*

Proof. Fix closed sets $C_s \subseteq \mathcal{N}_s \times \mathbb{N}^{\mathbb{N}}$ and continuous parameter-preserving functions $\phi_s\colon C_s \to \kappa^{\mathbb{N}} \times X$ such that $A_s = \phi_s(C_s)$. If $n \in \mathbb{N}$, then $C_n = \bigcup_{s\in\kappa^n} C_s$ is closed and $\phi_n = \bigcup_{s\in\kappa^n} \phi_s$ is continuous, so $\bigcup_{n\in\mathbb{N}} \phi_n(C_n)$ has uniformly analytic sections by Proposition 1.1. ⊠

Proposition 1.4. *Suppose that κ is an aleph. Then every quasi-Borel subset of $\kappa^{\mathbb{N}} \times \mathbb{N}^{\mathbb{N}}$ has uniformly analytic sections.*

Proof. The closed subsets of $\kappa^{\mathbb{N}} \times \mathbb{N}^{\mathbb{N}}$ clearly have uniformly analytic sections. As metrizability ensures that the open sets are countable unions of closed sets, Proposition 1.1 ensures that every open subset of $\kappa^{\mathbb{N}} \times \mathbb{N}^{\mathbb{N}}$ has uniformly analytic sections.

By Proposition 1.1 and induction, it is sufficient to show that if $(U_i)_{i\in I}$ is a pairwise disjoint sequence of open subsets of $\kappa^{\mathbb{N}} \times \mathbb{N}^{\mathbb{N}}$ and $B_i \subseteq U_i$ has uniformly analytic sections for all $i \in I$, then so too does $B = \bigcup_{i\in I} B_i$, as this implies that if $U_i \setminus B_i$ has uniformly analytic sections for all $i \in I$, then so too does $\sim B = \sim\bigcup_{i\in I} U_i \cup \bigcup_{i\in I} U_i \setminus B_i$.

For all $t \in \mathbb{N}^{<\mathbb{N}}$, note that the restriction of $\mathrm{proj}_{\kappa^{<\mathbb{N}}}$ to $I_t = \{(i,s) \in I \times \kappa^{<\mathbb{N}} \mid \mathcal{N}_s \times \mathcal{N}_t \subseteq U_i\}$ is injective. Propositions 1.1 and 1.3 ensure that $B = \bigcup_{t\in\mathbb{N}^{<\mathbb{N}}} \bigcup_{(i,s)\in I_t} B_i \cap (\mathcal{N}_s \times \mathcal{N}_t)$ has uniformly analytic sections. ⊠

Proposition 1.5. *Suppose that κ is an aleph, X is a Hausdorff space, and $A \subseteq \kappa^{\mathbb{N}} \times X$ has uniformly analytic sections. Then every relatively quasi-Borel subset of A has uniformly analytic sections.*

Proof. Fix a closed set $C \subseteq \kappa^{\mathbb{N}} \times \mathbb{N}^{\mathbb{N}}$ and a continuous parameter-preserving function $\phi\colon C \to \kappa^{\mathbb{N}} \times X$ such that $A = \phi(C)$. If $B \subseteq A$ is relatively quasi-Borel, then $\phi^{-1}(B)$ is relatively quasi-Borel in C. Propositions 1.2 and 1.4 therefore ensure that $B = \phi(\phi^{-1}(B))$ has uniformly analytic sections. ⊠

2. Baire category

We direct the reader to [Kec95, §8] for the basic facts concerning Baire category used below. Although $\kappa^{\mathbb{N}}$ is not separable when κ is uncountable, localization holds in every completely metrizable space.

Proposition 2.1. *Suppose that κ is an aleph, X is a topological space, and $B \subseteq \kappa^{\mathbb{N}} \times X$ is quasi-Borel. Then every horizontal section of B has the Baire property.*

Proof. As the horizontal sections of B are quasi-Borel, this follows from the fact that the family of subsets of $\kappa^{\mathbb{N}}$ with the Baire property contains the open sets and is closed under complements, countable unions, and open-separated unions. ⊠

3. Separation

Recall that $B \subseteq X$ *separates* $A_0, A_1 \subseteq X$ if $A_0 \subseteq B$ and $A_1 \cap B = \emptyset$.

Proposition 3.1. *Suppose that κ is an aleph, X is a Hausdorff space, and $A_0, A_1 \subseteq \kappa^{\mathbb{N}} \times X$ are disjoint sets with uniformly analytic sections. Then they are separated by a quasi-Borel set.*

Proof. Fix closed sets $C_i \subseteq \kappa^{\mathbb{N}} \times \mathbb{N}^{\mathbb{N}}$ and continuous parameter-preserving functions $\phi_i \colon C_i \to \kappa^{\mathbb{N}} \times X$ such that $A_i = \phi_i(C_i)$, and define $A_{i,s,t} = \phi_i(C_i \cap (\mathcal{N}_s \times \mathcal{N}_t))$ for all $i < 2$, $s \in \kappa^{<\mathbb{N}}$, and $t \in \mathbb{N}^{<\mathbb{N}}$.

Following the usual convention, we identify $\bigcup_{n \in \mathbb{N}} \kappa^n \times \mathbb{N}^n \times \mathbb{N}^n$ with $(\kappa \times \mathbb{N} \times \mathbb{N})^{<\mathbb{N}}$. Set $T = \{(s,t,u) \in (\kappa \times \mathbb{N} \times \mathbb{N})^{<\mathbb{N}} \mid A_{0,s,t} \cap \overline{A_{1,s,u}} \neq \emptyset\}$.

Lemma 3.2. *The tree T is well-founded.*

Proof. Suppose, towards a contradiction, that $(\sigma, c_0, c_1) \in [T]$. If $i < 2$, then $(\sigma, c_i) \in C_i$, so we can set $x_i = (\mathrm{proj}_X \circ \phi_i)(\sigma, c_i)$. As A_0 and A_1 are disjoint, it follows that $x_0 \neq x_1$, so there are disjoint open neighborhoods U_0 and U_1 of x_0 and x_1. Fix $n \in \mathbb{N}$ such that $A_{i,\sigma\restriction n, c_i\restriction n} \subseteq \kappa^{\mathbb{N}} \times U_i$ for all $i < 2$. Then $A_{0,\sigma\restriction n, c_0\restriction n} \cap \overline{A_{1,\sigma\restriction n, c_1\restriction n}} = \emptyset$, which contradicts the fact that $(\sigma \restriction n, c_0 \restriction n, c_1 \restriction n) \in T$. ⊠

Set $S_s = \{s \frown (\alpha) \mid \alpha < \kappa\}$ and recursively define by

$$B_{s,t,u} = \begin{cases} \bigcup_{r \in S_s} (\mathcal{N}_r \times X) \cap \bigcup_{m \in \mathbb{N}} \bigcap_{n \in \mathbb{N}} B_{r, t \frown (m), u \frown (n)} & \text{if } (s,t,u) \in T, \\ \sim \overline{A_{1,s,u}} & \text{otherwise.} \end{cases}$$

The obvious induction shows that $B_{s,t,u}$ is quasi-Borel and separates $A_{0,s,t}$ from $A_{1,s,u}$, so $B_{\emptyset,\emptyset,\emptyset}$ is a quasi-Borel set separating A_0 from A_1. ⊠

Given $A \subseteq \kappa^{\mathbb{N}} \times X$ and $G \subseteq \kappa^{\mathbb{N}} \times (X \times X)$, we say that A is *G-independent* if $\forall \sigma \in \kappa^{\mathbb{N}}\ (A_\sigma \times A_\sigma) \cap G_\sigma = \emptyset$.

Proposition 3.3. *Suppose that κ is an aleph, X is a Hausdorff space, the sets $A \subseteq \kappa^{\mathbb{N}} \times X$ and $G \subseteq \kappa^{\mathbb{N}} \times (X \times X)$ have uniformly analytic sections, and A is G-independent. Then there is a quasi-Borel G-independent set $B \supseteq A$.*

Proof. By the proof of [dRM21, Proposition 1.2], using Proposition 3.1 in place of Lusin's separation theorem. ⊠

4. The uniform $\mathbb{G}_0$ dichotomy

The proof follows that of [dRM21, Theorem 1.4]. The principal difference is that terminality must be localized to basic open subsets of the parameter space. We therefore keep track of pairs consisting of an approximation and a finite parameter sequence and assemble the resulting independent sets via open-separated unions.

Proof of Theorem 1. Define $p_i\colon \kappa^{\mathbb{N}} \times (X \times X) \to \kappa^{\mathbb{N}} \times X$ by setting $p_i(\sigma, (x_0, x_1)) = (\sigma, x_i)$ for all $i < 2$, $\sigma \in \kappa^{\mathbb{N}}$, and $x_0, x_1 \in X$. As Propositions 1.1 and 1.2 ensure that $\bigcup_{i<2} p_i(G)$ has uniformly analytic sections, there are closed sets $C, D \subseteq \kappa^{\mathbb{N}} \times \mathbb{N}^{\mathbb{N}}$ and continuous parameter-preserving functions $\phi\colon C \to \kappa^{\mathbb{N}} \times X$ and $\psi\colon D \to \kappa^{\mathbb{N}} \times (X \times X)$ such that $\bigcup_{i<2} p_i(G) = \phi(C)$ and $G = \psi(D)$.

An *approximation* is a triple $a = (n^a, \phi^a, (\psi^a_k)_{k<n^a})$, where $n^a \in \mathbb{N}$, $\phi^a\colon 2^{n^a} \to \mathbb{N}^{n^a}$, and $\psi^a_k\colon 2^{n^a-(k+1)} \to \mathbb{N}^{n^a}$ for all $k < n^a$. An approximation b is a *one-step extension* of a if:

(1) $n^b = n^a + 1$,
(2) $\forall u \in 2^{n^a} \forall v \in 2^{n^b}\ (u \sqsubseteq v \implies \phi^a(u) \sqsubseteq \phi^b(v))$, and
(3) $\forall k < n^a \forall u \in 2^{n^a-(k+1)} \forall v \in 2^{n^b-(k+1)}\ (u \sqsubseteq v \implies \psi^a_k(u) \sqsubseteq \psi^b_k(v))$.

Given $\sigma \in \kappa^{\mathbb{N}}$, a *$\sigma$-configuration* is a triple $\gamma = (n^\gamma, \phi^\gamma, (\psi^\gamma_k)_{k<n^\gamma})$, where $\phi^\gamma\colon 2^{n^\gamma} \to \mathbb{N}^{\mathbb{N}}$ and $\psi^\gamma_k\colon 2^{n^\gamma-(k+1)} \to \mathbb{N}^{\mathbb{N}}$ for all $k < n^\gamma$, such that:

(1) $\forall u \in 2^{n^\gamma}\ \phi^\gamma(u) \in C_\sigma$,
(2) $\forall k < n^\gamma \forall v \in 2^{n^\gamma-(k+1)}\ \psi^\gamma_k(v) \in D_\sigma$, and
(3) $\forall k < n^\gamma \forall v \in 2^{n^\gamma-(k+1)}$
$$(\mathrm{proj}_{X\times X} \circ \psi_\sigma \circ \psi^\gamma_k)(v) = ((\mathrm{proj}_X \circ \phi_\sigma \circ \phi^\gamma)(\mathbb{s}_k \frown (i) \frown v))_{i<2}.$$

We say that γ is *compatible* with $Y \subseteq X$ if $(\mathrm{proj}_X \circ \phi_\sigma \circ \phi^\gamma)(2^{n^\gamma}) \subseteq Y$, and *compatible* with a if:

(1) $n^\gamma = n^a$,
(2) $\forall u \in 2^{n^a}\ \phi^a(u) \sqsubseteq \phi^\gamma(u)$, and
(3) $\forall k < n^a \forall v \in 2^{n^a-(k+1)}\ \psi^a_k(v) \sqsubseteq \psi^\gamma_k(v)$.

Given a set $Z \subseteq \kappa^{\mathbb{N}} \times X$, we say that a is *Z-terminal* at σ if no σ-configuration is compatible with Z_σ and a one-step extension of a. Let $A(a, Z)$ denote the set of pairs of the form $(\sigma, (\mathrm{proj}_X \circ \phi_\sigma \circ \phi^\gamma)(\mathbb{s}_{n^a}))$, where $\sigma \in \kappa^{\mathbb{N}}$ and γ is a σ-configuration compatible with a and Z_σ.

Lemma 4.1. *Suppose that a is an approximation, $\sigma \in \kappa^{\mathbb{N}}$, $Z \subseteq \kappa^{\mathbb{N}} \times X$, and a is Z-terminal at σ. Then $A(a, Z)_\sigma$ is G_σ-independent.*

Proof. By the proof of [dRM21, Lemma 1.5]. ⊠

Lemma 4.2. *If a is an approximation and $B \subseteq \kappa^{\mathbb{N}} \times X$ is quasi-Borel, then $A(a, B)$ has uniformly analytic sections.*

Proof. The set of pairs (σ, γ) for which γ is a σ-configuration compatible with a is closed, so the set of such pairs that are compatible with B_σ is quasi-Borel, thus Propositions 1.2 and 1.5 yield the desired result. ⊠

We say that a is *Z-terminal* on a set $U \subseteq \kappa^{\mathbb{N}}$ if it is Z-terminal at σ for all $\sigma \in U$. Set $A_s(a, Z) = A(a, Z) \cap (\mathcal{N}_s \times X)$. If $B \subseteq \kappa^{\mathbb{N}} \times X$ is quasi-Borel and a is B-terminal on $\mathcal{N}_s$, then Proposition 3.3 and Lemmas 4.1 and 4.2 yield a G-independent quasi-Borel set $A_s(a, B) \subseteq B_s(a, B) \subseteq \mathcal{N}_s \times X$.

Let $\mathscr{A}$ be the set of all approximations. We will recursively construct a decreasing sequence $(B^\alpha)_{\alpha < \kappa^+}$ of quasi-Borel subsets of $\kappa^{\mathbb{N}} \times X$ and an increasing sequence $(S^\alpha)_{\alpha < \kappa^+}$ of subsets of $\kappa^{<\mathbb{N}} \times \mathscr{A}$, starting with $S^0 = \emptyset$ and setting $S^\delta = \bigcup_{\alpha < \delta} S^\alpha$ for all limit ordinals $\delta < \kappa^+$. Given $\alpha < \kappa^+$, $(B^\beta)_{\beta < \alpha}$, and $(S^\beta)_{\beta \leq \alpha}$, define $\rho^\alpha \colon S^\alpha \to \alpha$ by $\rho^\alpha(s, a) = \min\{\beta < \alpha \mid (s, a) \in S^{\beta+1}\}$, set $B^\alpha = {\sim}\bigcup_{a \in \mathscr{A}} \bigcup_{s \in \kappa^{n^a} \cap (S^\alpha)^a} B_s(a, B^{\rho^\alpha(s,a)})$ and $S^{\alpha+1} = S^\alpha \cup \{(s, a) \in \kappa^{<\mathbb{N}} \times \mathscr{A} \mid |s| = n^a \text{ and } a \text{ is } B^\alpha\text{-terminal on } \mathcal{N}_s\}$, and fix $B_s(a, B^\alpha)$ as above for all $(s, a) \in S^{\alpha+1} \setminus S^\alpha$. As $|\kappa^{<\mathbb{N}} \times \mathscr{A}| = \kappa$, there is a least $\lambda < \kappa^+$ with $S^\lambda = S^{\lambda+1}$, in which case $B^\lambda = B^{\lambda+1}$. Set $B = B^\lambda$, $S = S^\lambda$, and $\rho = \rho^\lambda$.

Lemma 4.3. *Suppose that $\alpha < \kappa^+$, a is an approximation, $s \in \kappa^{n^a}$, and a is not $B^{\alpha+1}$-terminal on $\mathcal{N}_s$. Then there exist $\xi < \kappa$ and a one-step extension b of a that is not B^α-terminal on $\mathcal{N}_{s^\frown(\xi)}$.*

Proof. Fix $\sigma \in \mathcal{N}_s$ for which a is not $B^{\alpha+1}$-terminal at σ. Then there is a σ-configuration γ compatible with a one-step extension b of a and $B^{\alpha+1}_\sigma$. Set $\xi = \sigma(n^a)$. If b is B^α-terminal on $\mathcal{N}_{s^\frown(\xi)}$ and $\beta = \rho^{\alpha+1}(s \frown (\xi), b)$, then $(\sigma, (\mathrm{proj}_X \circ \phi_\sigma \circ \phi^\gamma)(\mathbb{s}_{n^b})) \in A_{s^\frown(\xi)}(b, B^\beta) \cap B^{\alpha+1} \subseteq B_{s^\frown(\xi)}(b, B^\beta) \cap B^{\alpha+1} = \emptyset$, the desired contradiction. ⊠

Let a_0 denote the unique approximation for which $n^{a_0} = 0$.

Lemma 4.4. *Suppose that a_0 is not B-terminal on $\kappa^{\mathbb{N}}$. Then there is a continuous homomorphism from $\mathbb{G}_0$ to G_σ for some $\sigma \in \kappa^{\mathbb{N}}$.*

Proof. By Lemma 4.3 there exist $a_n \in \mathscr{A}$ with $n^{a_n} = n$, $s_n \in \kappa^n$, $\sigma_n \in \mathcal{N}_{s_n}$, and a σ_n-configuration γ_n compatible with a_{n+1} and B_{σ_n}, where a_{n+1} is a one-step extension of a_n and $s_n \sqsubseteq s_{n+1}$, for all $n \in \mathbb{N}$.

Set $\sigma = \bigcup_{n \in \mathbb{N}} s_n$. Then $\sigma_n \to \sigma$. Define $\phi' \colon 2^{\mathbb{N}} \to \mathbb{N}^{\mathbb{N}}$ and $\psi'_k \colon 2^{\mathbb{N}} \to \mathbb{N}^{\mathbb{N}}$ by $\phi'(c) = \bigcup_{n \in \mathbb{N}} \phi^{a_n}(c \restriction n)$ and $\psi'_k(c) = \bigcup_{n > k} \psi^{a_n}_k(c \restriction (n - (k+1)))$. If $c \in 2^{\mathbb{N}}$ and $k \in \mathbb{N}$, then $(\sigma_n, \phi^{\gamma_n}(c \restriction n+1)) \to (\sigma, \phi'(c))$ and

$(\sigma_n, \psi_k^{\gamma_n}(c \restriction (n-k))) \to (\sigma, \psi'_k(c))$, so the fact that C and D are closed ensures that $(\sigma, \phi'(c)) \in C$ and $(\sigma, \psi'_k(c)) \in D$.

Define $\pi\colon 2^{\mathbb{N}} \to X$ by $\pi(c) = (\mathrm{proj}_X \circ \phi_\sigma)(\phi'(c))$ and observe that $(\mathrm{proj}_{X\times X} \circ \psi_\sigma)(\psi'_k(c)) = (\pi(\mathbb{s}_k \frown (i) \frown c))_{i<2}$ for all $k \in \mathbb{N}$ and $c \in 2^{\mathbb{N}}$, so π is a continuous homomorphism from $\mathbb{G}_0$ to G_σ. ⊠

By Lemma 4.4, we can assume that a_0 is B-terminal on $\kappa^{\mathbb{N}}$. Set $\alpha = \rho(\emptyset, a_0)$. Then $A(a_0, B) = B \cap \bigcup_{i<2} p_i(G)$ and $A(a_0, B^\alpha) \subseteq B_\emptyset(a_0, B^\alpha)$, so $A(a_0, B) = \emptyset$, thus B is G-independent. For all $a \in \mathscr{A}$, define $S_a = \{s \in \kappa^{n^a} \mid (s, a) \in S\}$ and observe that the quasi-Borel set $B_a = \bigcup_{s\in S_a} B_s(a, B^{\rho(s,a)})$ is G-independent, so (1) follows from the fact that $\mathscr{A}$ is countable and $\kappa^{\mathbb{N}} \times X = B \cup \bigcup_{a\in\mathscr{A}} B_a$.

It only remains to note that if both (1) and (2) hold, then there is a quasi-Borel $\mathbb{N}$-coloring of $\mathbb{G}_0$. As $2^{\mathbb{N}}$ is second countable, every such function is Borel, contradicting [dRM21, Proposition 1.3]. ⊠

5. Stratifications and category transforms

For all sets $B \subseteq \kappa^{\mathbb{N}} \times X$ and sequences $t \in \kappa^{<\mathbb{N}}$, define

$$B^{*t} = \{x \in X \mid B^x \text{ is comeager in } \mathcal{N}_t\}.$$

Proposition 5.1. *Suppose that κ is an aleph, $t \in \kappa^{<\mathbb{N}}$, X is a κ-Souslin space, and $B \subseteq \kappa^{\mathbb{N}} \times X$ is quasi-Borel. Then B^{*t} is κ^+-Borel.*

Proof. By induction on the construction of B. If $B = U$ is open, then set $U_v = \bigcup\{V \subseteq X \mid V \text{ is open and } \mathcal{N}_v \times V \subseteq U\}$ and observe that $U^{*t} = \bigcap_{u\sqsupseteq t} \bigcup_{v\sqsupseteq u} U_v$ is κ^+-Borel. If B^{*t} is κ^+-Borel for all $t \in \kappa^{<\mathbb{N}}$, then so too is $(\sim B)^{*t} = \sim\bigcup_{u\sqsupseteq t} B^{*u}$ for all $t \in \kappa^{<\mathbb{N}}$. If B_i^{*t} is κ^+-Borel for all $i \in I$ and $t \in \kappa^{<\mathbb{N}}$ and $B = \bigcup_{i\in I} B_i$ is a countable or open-separated union, then the fact that X is κ-Souslin ensures that $|I| \leq \kappa$, and localization implies that $B^{*t} = \bigcap_{u\sqsupseteq t} \bigcup_{v\sqsupseteq u} \bigcup_{i\in I} B_i^{*v}$. ⊠

Given a closed set $C \subseteq \kappa^{\mathbb{N}}$ and a continuous function $\phi\colon C \to X$, define $C' = \{(\sigma, b) \in \kappa^{\mathbb{N}} \times \mathbb{N}^{\mathbb{N}} \mid \sigma \circ b \in C\}$ and $\phi'\colon C' \to \kappa^{\mathbb{N}} \times X$ by $\phi'(\sigma, b) = (\sigma, \phi(\sigma \circ b))$. A *stratification* of a κ-Souslin set $A \subseteq X$ is a set $A' \subseteq \kappa^{\mathbb{N}} \times X$ for which there exist a closed set $C \subseteq \kappa^{\mathbb{N}}$ and a continuous function $\phi\colon C \to X$ such that $A = \phi(C)$ and $A' = \phi'(C')$.

Proposition 5.2. *Suppose that κ is an aleph, X is a Hausdorff space, $A \subseteq X$ is κ-Souslin, $A' \subseteq \kappa^{\mathbb{N}} \times X$ is a stratification of A, and $x \in A$. Then $\forall^*\sigma \in \kappa^{\mathbb{N}}\ x \in A'_\sigma$.*

Proof. Fix a closed set $C \subseteq \kappa^{\mathbb{N}}$ and a continuous function $\phi\colon C \to X$ with the property that $A = \phi(C)$ and $A' = \phi'(C')$, as well as $\tau \in C$ for

which $\phi(\tau) = x$. Then the set $H = \bigcap_{n\in\mathbb{N}} \bigcup_{s\in\kappa^{<\mathbb{N}}} \mathcal{N}_{s^\frown(\tau(n))}$ is a dense G_δ subset of $\kappa^\mathbb{N}$ and $\forall \sigma \in H\ x \in A'_\sigma$. ⊠

Proposition 5.3. *Suppose that κ is an aleph, $t \in \kappa^{<\mathbb{N}}$, X is a Hausdorff space, G is a κ-Souslin digraph on X, G' is a stratification of G, and $B \subseteq \kappa^\mathbb{N} \times X$ is G'-independent. Then B^{*t} is G-independent.*

Proof. Otherwise, fix $x, y \in B^{*t}$ with $x\ G\ y$ and appeal to Proposition 5.2 to obtain $\sigma \in \mathcal{N}_t$ with $x, y \in B_\sigma$ and $x\ G'_\sigma\ y$, a contradiction. ⊠

Proof of Theorem 2. Fix a stratification G' of G. By Theorem 1 we can assume that there is a quasi-Borel function $c' \colon \kappa^\mathbb{N} \times X \to \mathbb{N}$ such that c'_σ is a coloring of G'_σ for every $\sigma \in \kappa^\mathbb{N}$. Set $B_n = c'^{-1}(\{n\})$.

Set $V = \{x \in X \mid \exists y \in X\ (x\ G\ y \text{ or } y\ G\ x)\}$. For all $n \in \mathbb{N}$ and $t \in \kappa^{<\mathbb{N}}$, Proposition 5.1 yields a κ^+-Borel set $C_{n,t} \subseteq X$ such that $C_{n,t} \cap V = B^{*t}_n \cap V$. Proposition 5.3 ensures that B^{*t}_n is G-independent, thus so too is $C_{n,t}$.

If $x \in V$, then $\kappa^\mathbb{N} = \bigcup_{n\in\mathbb{N}} B^x_n$, so some B^x_n is non-meager. Proposition 2.1 and localization yield $t \in \kappa^{<\mathbb{N}}$ with $x \in B^{*t}_n$, so $x \in C_{n,t}$, thus $C_* = X \setminus \bigcup_{n\in\mathbb{N},t\in\kappa^{<\mathbb{N}}} C_{n,t}$ is κ^+-Borel and disjoint from V, hence G-independent. As C_* and $C_{n,t}$ form a cover of X by at most κ-many κ^+-Borel G-independent sets, there is a κ^+-Borel coloring of G. □

As noted in [Mil12, §8], the weakening of Theorem 2 without the definability condition on c follows from the simplification of the classical proof of the $\mathbb{G}_0$ dichotomy in which the use of separation is removed. As separation easily generalizes to κ-Souslin sets, this suggests a simpler route towards a classical proof of the full version of Theorem 2. The direct adaptation of that proof, however, would require separation for continuous images of κ^+-Borel sets. To see that this is consistently false, note that every $\mathbf{\Pi}^1_2$ subset of $\mathbb{N}^\mathbb{N}$ is $\aleph_2$-Borel (see, for example, [Kec95, Exercise 38.8]), so every $\mathbf{\Sigma}^1_3$ subset of $\mathbb{N}^\mathbb{N}$ is a continuous image of an $\aleph_2$-Borel set. Under $\mathtt{MA}+\neg\mathtt{CH}$, the sets with the Baire property are closed under $\aleph_1$-unions, so every $\aleph_2$-Borel subset of $\mathbb{N}^\mathbb{N}$ has the Baire property. But under $\mathtt{MA} + \neg\mathtt{CH} + \omega_1 = \omega^L_1$, there is a $\mathbf{\Delta}^1_3$ set $B \subseteq \mathbb{N}^\mathbb{N}$ without the Baire property by [HS85, Claim 8], so B is not $\aleph_2$-Borel, thus B and $\sim B$ are continuous images of $\aleph_2$-Borel sets.

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